

HTLV-III/LAV directly interacts with the T4 molecule on the target cell (22). This also suggests that conserved epitopes of the *env* proteins of these viruses may be critical to virus infection and thus appropriate candidates for vaccine development.

The occurrence of an STLV-III_{AGM}-related virus in healthy Senegalese may provide clues as to how this family of viruses has evolved. It is well recognized that HTLV-III/LAV shares some sequence homology with various ungulate retroviruses, most notably visna (23). However, the degree of homology in genes other than *pol* is negligible, suggesting a more ancestral relationship. The similarities and cross-reactivity of STLV-III_{AGM} with the *env*, *gag*, *pol*, and 3' *orf* products of HTLV-III/LAV indicate that these are more closely related members of the family of T-lymphotropic viruses. STLV-III_{AGM} naturally infects only a limited number of primate species, unlike STLV-I, suggesting that STLV-III_{AGM} may have arisen in more recent times. It is therefore conceivable that STLV-III_{AGM} or HTLV-IV may have served as the progenitor virus to the human AIDS virus, HTLV-III/LAV; alternatively they may have had a common progenitor. Continued study of the origin of

this group of viruses may provide a better understanding of how they acquired their pathogenicity and how we can effectively interfere with HTLV-III/LAV infection.

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25. We thank M. Larned, C. S. Boye, and A. Gaye for technical assistance, M. G. Sarngadharan for monoclonal antibody to HTLV-III p24, and J. Groopman for human reference sera from patients in the United States with AIDS and ARC. This research was supported by NIH research grants CA 37466 and CA 18216. P.J.K. and J.S.A. were supported by National Research Service Award ST 52 CA9382 and F.B. by F. Rabelais University and the Association pour le Recherche sur le Cancer (Villejuif). T.H.L. is a Special Fellow of the Leukemia Society of America.

31 January 1986; accepted 17 March 1986

El Niño: A Chaotic Dynamical System?

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Most of the principal qualitative features of the El Niño–Southern Oscillation phenomenon can be explained by a simple but physically motivated theory. These features are the occurrence of sea-surface warmings in the eastern equatorial Pacific and the associated trade wind reversal; the aperiodicity of these events; the preferred onset time with respect to the seasonal cycle; and the much weaker events in the Atlantic and Indian oceans. The theory, in its simplest form, is a conceptual model for the interaction of just three variables, namely near-surface temperatures in the east and west equatorial ocean and a wind-driven current advecting the temperature field. For a large range of parameters, the model is naturally chaotic and aperiodically produces El Niño-like events. For a smaller basin, representing a smaller ocean, the events are proportionally less intense.

chaotic dynamic or probabilistic models (12). To forecast El Niño, a coupled general circulation model is probably needed. However, to understand the phenomenon a simpler theory is needed. The theory, or model, presented here differs from others in that no stochastic or seasonally varying forcing is required to produce the aperiodicity, no explicit wave dynamics are required to explain the time scales, and no mid-latitude influences are needed to explain variations in intensity. Instead, these phenomena arise naturally and deterministically within a simple framework. It has been conjectured (6, 10, 13) that El Niños may be generated internally, but no mechanism for the com-

EL NIÑO MAY BE DEFINED AS THE appearance of anomalously warm water in the eastern equatorial Pacific. Associated with this is a weakening, and sometimes a reversal, of the trade wind field (Fig. 1). Major El Niño–Southern Oscillation events occurred in 1957, 1965, 1972, and 1982, and their occurrence for the past 100 years can be inferred from proxy data (1). The various events differ in detail and intensity but appear to have broadly similar overall features (2–4). Although occurring aperiodically, the events also appear to be at

least partially phase-locked to the seasonal cycle. As well as engendering much purely scientific interest, El Niño has major economic consequences and possibly global climatic effects (5). Much observational and modeling effort has therefore been devoted to it, with various degrees of success (6–12).

In this report I show that all the broad qualitative features may be explained with a simple but realistic model. Other models for El Niño exist, varying in complexity from relatively elaborate coupled ocean-atmosphere models (10) to relatively simple sto-

plete cycle of events has been isolated.

Imagine an equatorial ocean to be a box of fluid characterized by temperatures in the east and west (T_e and T_w) and a current u (Fig. 2). The current is driven by a surface wind, which is in part generated by the temperature gradient $(T_e - T_w)/\Delta x$, so describing a parameterized Walker circulation (14). A cooler temperature in the east ($T_e < T_w$) produces a westward surface

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